

Simulation of heat diffusion with Crank-Nicolson method

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24th November 2025

1 Introduction

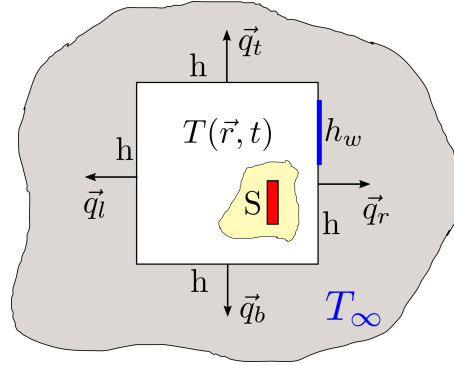


Figure 1: Geometry of simulation box. The room exchange the heat with the heat bath of constant temperature T_∞ through the walls and a window on right wall (blue). The room has a heater (red color) supplying the energy at a constant rate when it is turned on. Heat can leak from the room due not perfect heat isolation what marks the fluxes $\vec{q}$.

The diffusion equation can be applied in simulation of heat conduction, then it can have the following form

$$\frac{\partial T}{\partial t} = D \nabla^2 T + S \quad (1)$$

In this equation D is a diffusion constant, the function $T(\vec{r}, t)$ describes the spatial-time distribution of temperature and $S(\vec{r}, t)$ is the source/sink term that may supply the heat to the system (heater) or take it away (cooler).

In project we will simulate numerically time variations in temperature distribution in the room featured with one electric heater, walls (heat conductors) and a window (heat conductor) [see Fig.1]. This problem is to be solved in 2D geometry on a mesh using finite difference method and Crank-Nicolson time integrator.

1.1 Geometry, mesh of nodes

The room has square geometry with edge length $\mathcal{L}$. As we intend to use finite difference method, we need to define the mesh of nodes before. The number of nodes in both x-y directions are the same, hence these are equidistant with the same spatial step $\Delta x = \Delta y = \Delta$, positions of nodes are set as

follows

$$\Delta = \frac{\mathcal{L}}{N-1} \quad (2)$$

$$x_i = \Delta \cdot i, \quad i = 0, 1, \dots, N-1 \quad (3)$$

$$y_j = \Delta \cdot j, \quad j = 0, 1, \dots, N-1 \quad (4)$$

Similarly we discretize the time domain

$$t = t_n = \Delta t \cdot n, \quad n = 0, 1, 2, \dots, \frac{t_{max}}{\Delta t} \quad (5)$$

The heater extends over the area $[x_a, x_b] \times [y_a, y_b]$ and in this region it supplies the heat at constant rate with homogeneous distribution

$$S(x, y) = \begin{cases} S_{max}, & \vec{r} \in [x_a, x_b] \times [y_a, y_b] \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

From now on we will use the quantities T and S defined on spatio-temporal mesh i.e. $T(x, y, t) \rightarrow T_{i,j}^n$ and $S(x, y, t) \rightarrow S_{i,j}^n$.

The window is located at the right wall and extends over $y_{window} \in [y_1, y_2]$. The window and the walls have the width equal to single spatial step Δ .

1.2 Boundary conditions

In order to solve the diffusion equation numerically we need to specify the boundary conditions (BC) at the walls and at window. These we derive based on comparison of the heat flux incident on the inner side of wall with the outgoing flux on the outer side, these BC must be met at every point on the room's boundary

$$\vec{q}_{in} = \vec{q}_{out} \quad (7)$$

$$\vec{q}_{in} = -D \left(\vec{n} \cdot \vec{\nabla} T \right) \vec{n} \quad (\text{diffusive flux}) \quad (8)$$

$$\vec{q}_{out} = h(T - T_\infty) \vec{n} \quad (\text{convective flux}) \quad (9)$$

By applying this condition to four walls (and a window) we get the following BCs

1. BC on west wall:

$$T_{0,j}^n = \frac{h\Delta}{D+h\Delta} T_\infty + \frac{D}{D+h\Delta} T_{1,j}^n, \quad i = 0, j = 1, 2, \dots, N-2 \quad (10)$$

2. BC on north wall:

$$T_{i,N-1}^n = \frac{h\Delta}{D+h\Delta} T_\infty + \frac{D}{D+h\Delta} T_{i,N-2}^n, \quad i = 1, 2, \dots, N-2, j = N-1 \quad (11)$$

3. BC on east wall:

$$T_{N-1,j}^n = \frac{h\Delta}{D+h\Delta} T_\infty + \frac{D}{D+h\Delta} T_{N-2,j}^n, \quad i = N-1, j = 1, 2, \dots, N-2 \quad (12)$$

4. BC on south wall:

$$T_{i,0}^n = \frac{h\Delta}{D+h\Delta} T_\infty + \frac{D}{D+h\Delta} T_{i,1}^n, \quad i = 1, 2, \dots, N-2, j = 1 \quad (13)$$

The four corners do not enter calculations and BC, therefore we set values at these points as averages of their neighbours

$$T_{0,0}^n = \frac{1}{2}(T_{0,1}^n + T_{1,0}^n) \quad (14)$$

$$T_{0,N-1}^n = \frac{1}{2}(T_{0,N-2}^n + T_{1,N-1}^n) \quad (15)$$

$$T_{N-1,N-1}^n = \frac{1}{2}(T_{N-2,N-1}^n + T_{N-1,N-2}^n) \quad (16)$$

$$T_{N-1,0}^n = \frac{1}{2}(T_{N-2,0}^n + T_{N-1,1}^n) \quad (17)$$

1.3 Crank-Nicolson method

CN is one time-step method in which the r.h.s of Eq.1 is taken at present n and next $n + 1$ time instants with the same weights $1/2$

$$\begin{aligned} \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} &= \frac{D}{2\Delta^2} (T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n - 4T_{i,j}^n) \\ &+ \frac{D}{2\Delta^2} (T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1}) \\ &+ \frac{1}{2} (S_{i,j}^n + S_{i,j}^{n+1}) \end{aligned} \quad (18)$$

It is plausible to separate time instant terms, keeping $n + 1$ -dependent terms on the left and n -dependent terms on the right

$$L_{i,j} = R_{i,j} \quad (19)$$

$$L_{i,j} = T_{i,j}^{n+1} - \frac{D\Delta t}{2\Delta^2} (T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1}) - \frac{\Delta t}{2} S_{i,j}^{n+1} \quad (20)$$

$$R_{i,j} = T_{i,j}^n + \frac{D\Delta t}{2\Delta^2} (T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n - 4T_{i,j}^n) + \frac{\Delta t}{2} S_{i,j}^n \quad (21)$$

here factors $L_{i,j}$ and $R_{i,j}$ mean that we calculate both sides at node (i, j) . $R_{i,j}$ is calculated at present time step so we know here everything. On the left side $L_{i,j}$ depends on the new solution at $n + 1$ time instant not at only one but at a few nodes, hence CN method is an implicit one. Structure $L_{i,j}$ resembles discretized Poisson equation, so we may try to solve it with *Gauss-Seidel relaxation*, then $T_{i,j}^{n+1}$ requires to be made dependent on all other factors

$$T_{i,j}^{n+1} = \frac{1}{1 + \frac{2D\Delta t}{\Delta^2}} \left[\frac{D\Delta t}{2\Delta^2} (T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1}) + \frac{1}{2} S_{i,j}^{n+1} + R_{i,j} \right] \quad (22)$$

1.3.1 Gauss-Seidel algorithm for one step in CN

To make one step in CN method we use Gauss-Seidel method, recall that it needs only single array to keep the old ($T_{i,j}^n$) and the upgraded ($T_{i,j}^{n+1}$) values, therefore following pseudocode for GS relaxation involves Eq.22 without time indices

$$T_{i,j} \leftarrow \frac{1}{1 + \frac{2D\Delta t}{\Delta^2}} \left[\frac{D\Delta t}{2\Delta^2} (T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}) + \frac{1}{2} S_{i,j} + R_{i,j} \right] \quad (23)$$

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initialize:  $K_{max}$ ,  $TOL$ 
calculate all elements:  $R_{i,j}$ 

for k=1 to  $K_{max}$  do

    for i=1 to N-2 do
        for j=1 to N-2 do
             $T_{i,j} \leftarrow Eq.23$ 
        end do
    end do

    apply boundary conditions: Eqs.10, 11, 12, 13
    upgrade values at corners: Eqs. 14, 15, 16, 17

    //check STOP condition – norm of residual vector
    c=0
    for i=1 to N-2 do
        for j=1 to N-2 do
             $L_{i,j} = Eq.20$  for upgraded  $T_{i,j}$ 
             $c = c + (L_{i,j} - R_{i,j})^2 \Delta^2$ 
        end do
    end do
    if (  $\sqrt{c} < TOL$  ) break
end do

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1.3.2 Automatic control of temperature

Now we have almost all elements at hand. Because the aim is to simulate the heating of room, we may add the last ingredient which is the control of temperature in the course of simulation. Therefore at the node $(x_c, y_c) \rightarrow (i_c, j_c)$ we put the temperature sensor which has two states: ON and OFF, the former make the heater starts to operate (turned on) when temperature drops below T_{low} while the later turns it off when it grows up to T_{high} . One may use this switcher in calculation by introducing new weighting factor which multiplies the source terms $w \cdot S_{i,j}^n$ and $w \cdot S_{i,j}^{n+1}$. Temporary state of switcher can be set as follows

$$if(T_{i_c, j_c}^n < T_{low}) \quad w = 1 \quad (24)$$

$$else \quad if(T_{i_c, j_c}^n > T_{high}) \quad w = 0 \quad (25)$$

1.3.3 Supplied and lost energy

When heater is turned on it supplies the energy to the room whilst the walls and the window give it away to the environment (heat bath). We may be interested in how much energy is temporary delivered (during one time step)

$$E_{supplied}(t_n) = \int_{t_n}^{t_{n+1}} dt \int_{room} d^2r w(t) \cdot S(x, y, t) \quad (26)$$

$$E_{supplied}^n = \frac{1}{2}(w + w_{old}) \cdot \Delta^2 \cdot \Delta t \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} S_{i,j}^n \quad (27)$$

and lost e.g. by the flux passing through the window

$$E_{window}(t_n) = \int_{t_n}^{t_{n+1}} dt \int_{window} dy q_w \quad (28)$$

$$E_{window}^n = \Delta \cdot \Delta t \sum_{j=j_1}^{j_2} h_w (T_{N-1,j}^n - T_\infty) \quad (29)$$

1.3.4 CN algorithm

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initialization:   $\mathcal{L}$ ,  $N$ ,  $\Delta$ ,  $\Delta t$ ,  $t_{max}$ ,  $n_{max} = \frac{t_{max}}{\Delta t}$ ,  $D$ ,
                 $h, h_w$ ,  $T_\infty$ ,
                 $w = 1$ ,  $T_{low}$ ,  $T_{high}$ ,  $(x_c, y_c) \rightarrow (i_c, j_c)$ 
                 $T_{i,j} \leftarrow T_\infty$  (all elements)
                 $t = 0$ 
for n=0 to  $n_{max}$  do

    (1) calculate all elements:  $R_{i,j} \leftarrow Eq.21$ 
    (2) save old weight w:  $w_{old} \leftarrow w$ 
    (3) change heater state if necessary: Eqs.24 and 25
    (4) make GS relaxation: algorithm Sec.1.3.1
    (5) calculate supplied and lost (window) energy: Eq.27 and Eq.29
    (6) change time:  $t = t + \Delta t$ 

    (7) save to file:  $t$ ,  $T_{i_c, j_c}$ ,  $E_{supplied}^n$ ,  $E_{window}^n$ 
    (8) save to another file (if needed): array  $T_{i,j}$ 

end do

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2 Practical part

Implement the CN algorithm for diffusion equation (Sec.1.3.4). Use the following parameters which are common for all simulations:

- simulation time step and number of steps: $\Delta t = 10$, $T_{max} = 10^4$,
- geometry: $\mathcal{L} = 10$, $N = 51$, $y_{window} \in [6, 9]$
- diffusion constant: $D = 0.1$,
- heater: $S(x, y) \in [8.0, 8.8] \times [2, 4]$, $S_{max} = 2$
- temperature sensor: position $(x_c, y_c) = (2, 8)$, lower threshold $T_{low} = 293$,
- temperature outside: $T_\infty = 273$,
- Gauss-Seidel relaxation: $k_{max} = 30$, $TOL = 10^{-8}$

Tasks to do

1. Set the same factors for walls (h) and window (h_w) $h = h_w = 0$ (perfect heat isolation), $T_{high} = 10^4$ (heater will be working constantly), perform simulation, then prepare the plot of sensor temperature T_{i_c, j_c} as function of time and make 2D heat maps for $t = 10^1, 10^2, 10^3, 10^4$.

2. Set $h = h_w = 0.002$ (the walls and window allows the heat flux move outside) and $T_{high} = 10^4$, perform simulation and make the same plots as in task 1.
3. Set $h = 0.002$ and $h_w = 0.01$ (the walls and window allow the heat flux flow outside, but the leakage of heat through window dominates) and $T_{high} = 10^4$, perform simulation and make the same plots as in task 1.
4. Set $h = 0.002$ and $h_w = 0.01$, $T_{high} = 298$ (reasonable threshold for highest temperature in the room), perform simulation and make the plot of sensor temperature T_{i_c, j_c} and four heat maps for some characteristic work points [see e.g. the full dots in top-right subfigure in Fig.2].
5. Calculations of supplied and lost heat energy.
Set $h = 0$ (perfect isolation at walls), $h_w = 1.0$ (broken window) and $T_{high} = 298$, perform simulation and compare the energies $E_{supplied}$ and E_{window} as functions of time in one figure.
6. Check the conservation energy law.
Set $h = 0$ (isolated walls), $h_w = 1.0$ (only window lets the heat flux go outside) and $T_{\infty} = 10^4$, perform simulation and compare the energies supplied (heater) and lost (window) as function of time in one figure.
7. At home prepare the report containing concise description of the problem, all generated results and comment on observed variations in heat maps, sensor temperature and energy supplied and lost for applied sets of initial parameters.

3 Example results

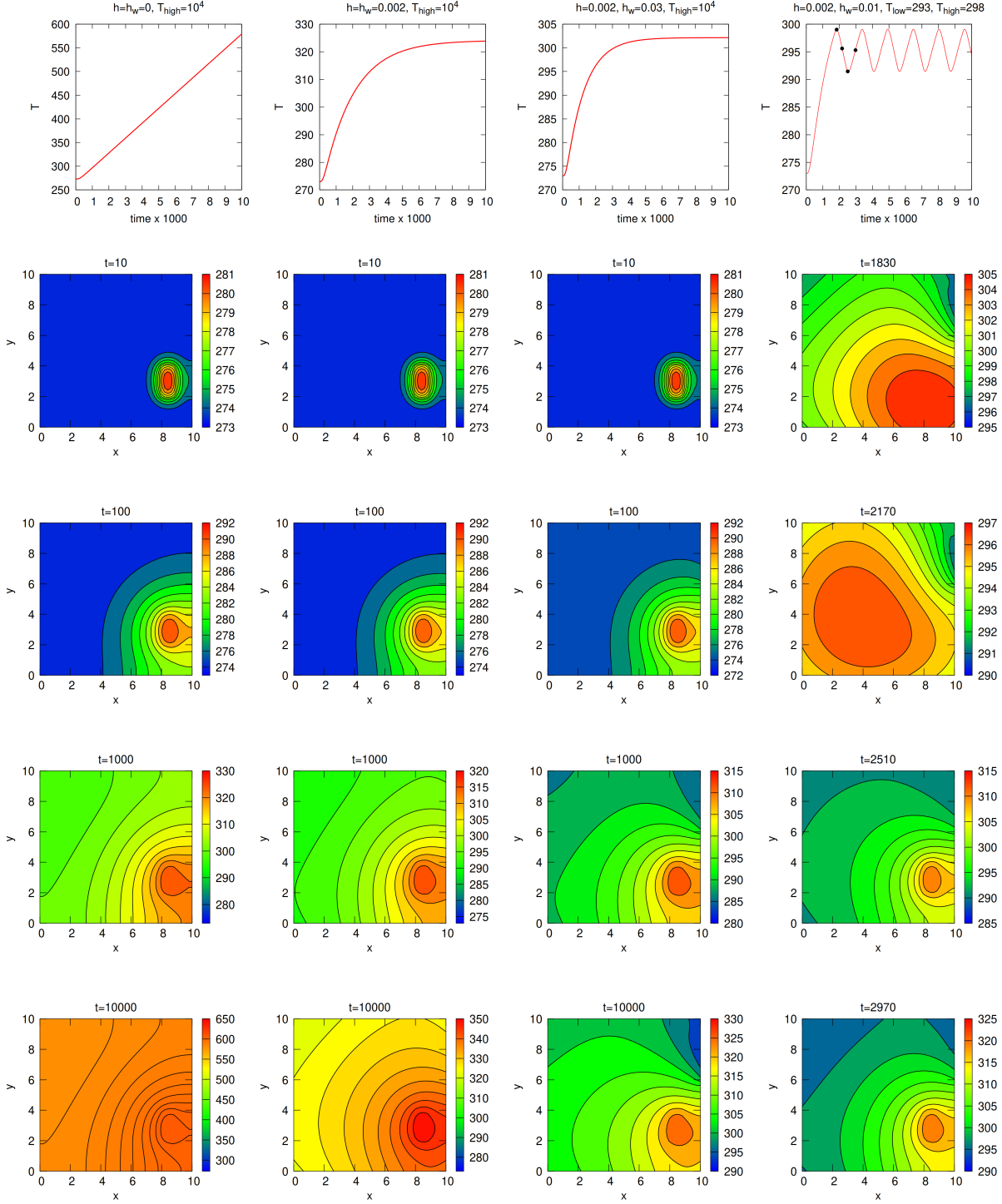


Figure 2: **Tasks 1-4.** The time variations of temperature at control point T_{i_c, j_c} and spatial temperature distribution in tasks 1-4 are shown in consecutive columns. In task 4 (4-th column) the four temperature distributions were registered at time instants marked with full dots in $T_{i_c, j_c}(t)$ plot (top-right subfigure).

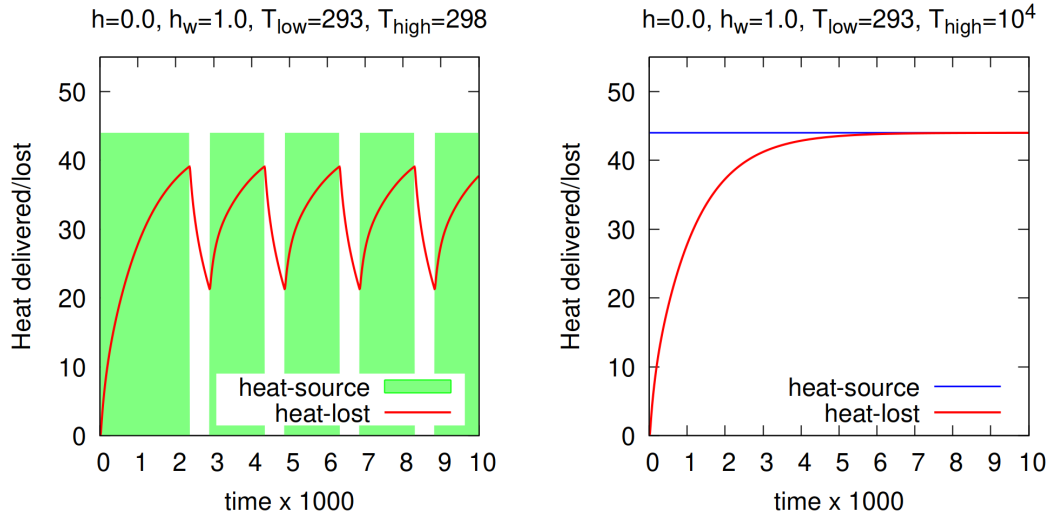


Figure 3: Left (Task 5): The rate of heat supplied by heater (green color) and heat lost due to window (red curve) as functions of time.

Right (Task 6): Heat supplied to system (blue curve) in continuous way and the heat lost due to defective window (red color). Notice that the walls are made perfect heat isolators, hence both fluxes, from the heater and flowing through the window must equilibrate after some time according to energy conservation law.